

Scalable Agentic Retrieval-Augmented Generation for Technical Audits

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1. Abstract & System Architecture

This paper introduces an end-to-end Agentic RAG framework designed for mission-critical document auditing. The architecture combines dense FAISS vector embeddings (MiniLM-L6) with sparse BM25 indexing via Reciprocal Rank Fusion (RRF with parameter $k=60$). In high-concurrency benchmarks, the system achieved a sub-200ms latency profile across 100,000 indexed records.

The system pipeline comprises three core micro-agents: the Planner Agent (which determines retrieval routing), the Synthesis Agent (which aggregates grounded document context), and the Self-RAG Fact-Checker (which verifies groundedness against source context to eliminate hallucinations).

2. Experimental Results & Performance Metrics

We evaluated the retrieval and classification framework across four standard evaluation datasets. The neural Cross-Encoder re-ranker achieved a Mean Reciprocal Rank (MRR@10) of 0.892, outperforming baseline vector search by 14.3%.

Model Architecture	Precision	Recall	F1-Score	Accuracy
Baseline BM25	78.4%	81.2%	79.7%	80.1%
Dense FAISS (MiniLM)	85.1%	88.6%	86.8%	86.5%
Hybrid (BM25 + Dense RRF)	91.3%	92.7%	92.0%	92.4%
Agentic Hybrid + Re-Ranker	95.8%	96.4%	96.1%	96.7%

Table 1: Quantitative comparison of retrieval-augmented classification strategies. As reported in Table 1, the Agentic Hybrid with Cross-Encoder achieved the highest overall accuracy of 96.7% and an F1-Score of 96.1%.

3. Technical Implementation & Deployment

The backend runtime is constructed using FastAPI and Uvicorn running on Python 3.11. Local vector stores utilize SQLite for relational metadata and FAISS IndexFlatIP for normalized inner-product similarity computation. For document ingestion, layout preservation is guaranteed via pdfplumber with pytesseract OCR fallback routines. The system is container-ready and deploys under standard Linux or Windows runtimes.